Year 12 Maths - Proof and mathematical communication

Similar questions - Proof by cases and deduction

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Prove that the square of any odd integer is of the form $8k+1$, where k is an integer. **[4]**

(b) Hence prove that n^2+7 is divisible by 8 for every odd integer n. **[2]**

2 Prove that n^3-n is divisible by 3 for every integer n, by considering separately the cases $n=3k$, $n=3k+1$ and $n=3k+2$. **[5]**

Question 1

(a) Let the odd integer be $2m+1$. $(2m+1)^2 = 4m^2+4m+1 = 4m(m+1)+1$. $m(m+1)$ is a product of consecutive integers, so it is even, say $2k$. Then the square is $8k+1$

(b) $n^2+7 = 8k+1+7 = 8(k+1)$, which is divisible by 8

Question 2

Factorise: $n^3-n = (n-1)n(n+1)$.

Case $n = 3k$: the factor $n = 3k$ is a multiple of 3.

Case $n = 3k+1$: the factor $n-1 = 3k$ is a multiple of 3.

Case $n = 3k+2$: the factor $n+1 = 3k+3 = 3(k+1)$ is a multiple of 3.

In every case one factor is a multiple of 3, so the product is divisible by 3 (these cases cover all integers)